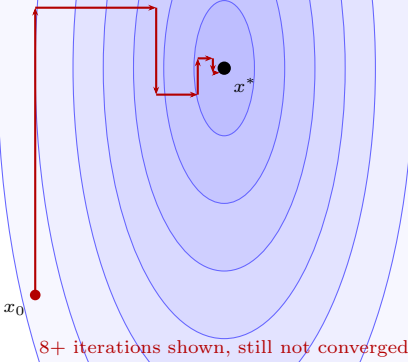


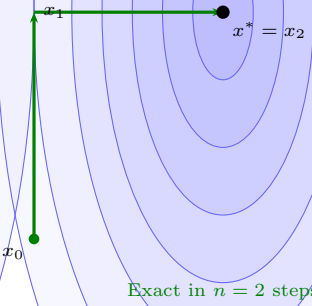
Steepest Descent vs. Conjugate Gradient

$$\min \frac{1}{2}x^T Ax - b^T x, \quad A = \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix}, \quad \kappa = 5$$

Steepest Descent



Conjugate Gradient

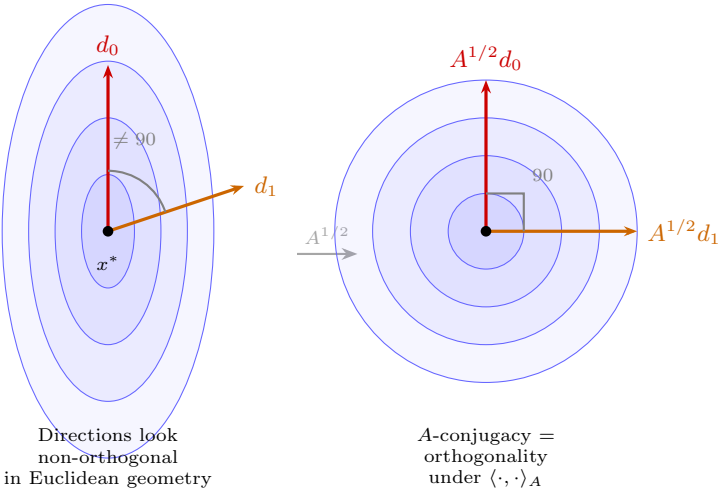


A-Conjugate Directions

$$d_i^T A d_j = 0: \text{orthogonal in the } A\text{-inner product}$$

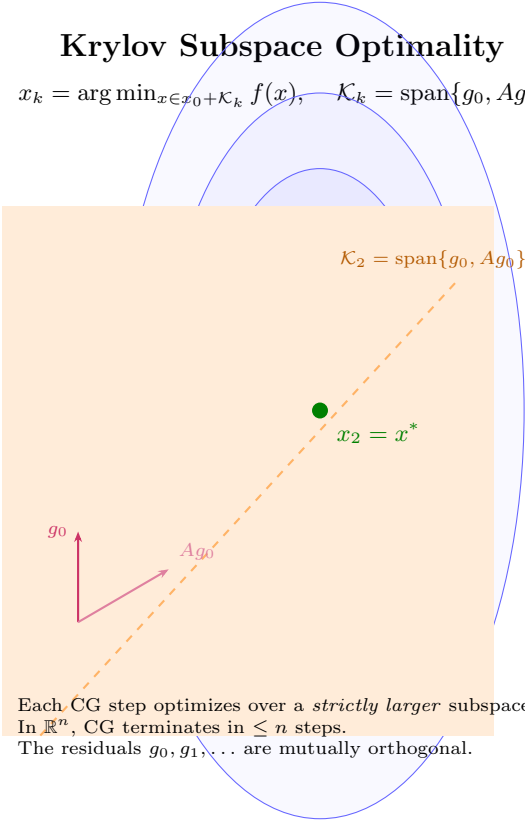
Original Space

A-Transformed Space



Krylov Subspace Optimality

$$\text{At step } k: \quad x_k = \arg \min_{x \in x_0 + \mathcal{K}_k} f(x), \quad \mathcal{K}_k = \text{span}\{g_0, Ag_0, \dots, A^{k-1}g_0\}$$



Convergence Rate: SD vs. CG

